

BUSINESS UNIT
School of Chemistry

DATE OF ASSESSMENT
2026-06-10

RA TITLE
Recrystallisation of benzoic acid

LOCATION
Joseph Black Building, Lab 2.14

DATE OF REVIEW
2027-06-10

RISK ASSESSOR
David Kemp

PERSONS AT RISK
Students

SOP REF NUMBER(S)
SOP-114

STEP	ACTIVITY	CHEMICALS	GHS SUMMARY	ENGINEERING	PPE	OTHER CONTROLS
Step 1	Dissolution and hot filtration Dissolve crude benzoic acid in minimum hot ethanol on a hotplate stirrer in the fume hood, then hot-filter.	Ethanol, Benzoic acid, Nitric acid		Fume hood	Lab coat, Goggles, Gloves	—

 Flammable

 Acute toxicity

 Corrosive

 Health hazard

 Oxidising

See [Appendix A — Chemical Detail](#) for full hazard statements, WELs, and exposure information per chemical.

03 TASK HAZARDS

HAZARD	HOW HARM OCCURS	BEFORE CONTROLS	CONTROLS IN PLACE	AFTER CONTROLS	FURTHER ACTION
Burns from hotplate and hot glassware	Contact with hot surfaces or hot solvent during dissolution and filtration; assessor and nearby students exposed.	L3 x S3 = 9	Heat-resistant gloves for handling hot glassware; hotplate positioned away from bench edge; allow apparatus to cool before handling.	L2 x S2 = 4	—

04 CONTROLS

Elimination / substitution

- Substitution was reviewed for ethanol and benzoic acid; no suitable lower-hazard alternative was identified for this task.

Reduction

- Only the quantity required of each chemical will be kept at the work area.

Administrative controls

- Work will be carried out in accordance with the approved SOP or local safe working procedure.

Monitoring / health

Air monitoring:

- Air monitoring is not required because existing controls are established and exposure is expected to remain below relevant WELs.

Health surveillance:

- Health surveillance is not required because no sensitiser, asthmagen, CMR or other health surveillance trigger has been identified.

05 STORAGE

Flammables Cabinet

Organic solvents and organic acids

Use for flammable organic solvents and flammable organic acids where SDS confirms this is appropriate.

Ethanol

Corrosives Cabinet

Non-oxidizing organic and mineral acids

Keep acids separate from bases; acids and bases should not share a cabinet.

Benzoic acid

Oxidizing acids in double containment

Double containment required; isolate from organic solvents/acids, bases and oxidizer cabinet contents.

Nitric acid

06 EMERGENCY RESPONSE AND WASTE

SPILLS	- For chemicals used in this task (Ethanol, Benzoic acid), local response is limited to small, contained spills only. Follow SDS Section 6 and the local spill plan for the detailed procedure.
FIRST AID	- Eye exposure with serious eye-damage hazards: rinse immediately using eyewash while holding eyelids open and seek urgent medical attention.
FIRE	- For chemicals used in this task (Ethanol, Benzoic acid), follow SDS Section 5 and the local fire/emergency plan for suitable extinguishing media, special hazards and firefighter precautions.
WASTE HANDLING	- Collect contaminated solid waste from this task (Benzoic acid) in the compatible solid-waste stream specified by the SDS/local procedure; do not use solid containers for liquid waste.
OTHER	—

07 BRIEFING & SIGN-OFF

NAME	DATE	SIGNATURE
Alex Murray	2026-06-10	

A1 APPENDIX A — CHEMICAL DETAIL

#	CHEMICAL	FORM / QTY	HAZARD STATEMENTS	WEL	STORAGE	EXPOSURE	GHS
1	Ethanol CAS 64-17-5	Forms: liquid Mass/volume range: 250 mL	H225 Highly Flammable liquid and vapor	TWA: n/a STEL: n/a Source: EH40	Organic solvents and organic acids incompatible absorbents: Cellulose-Based Absorbents Source: GHS/PubChem + CAMEO Confidence: High GHS flammable H-code or pictogram. CAMEO groups: 4: Alcohols and Polyols H-codes: H225 GHS pictograms: flammable PubChem flash point: 12.8 °C PubChem vapour pressure: 5.9 kPa PubChem physical form: liquid PubChem physical description: Ethanol with a small amount of an adulterant added so as to be unfit for use as a beverage.	30 min weekly skin	
2	Benzoic acid CAS 65-85-0	Forms: solid Mass/volume range: 50 g	H315 Causes skin irritation; H318 Causes serious eye damage; H372 Causes damage to organs through prolonged or repeated exposure	TWA: n/a STEL: n/a Source: Manual	Non-oxidizing organic and mineral acids separate from bases; retain toxic-label controls and secondary containment Source: GHS/PubChem + CAMEO Confidence: High CAMEO acid reactive group. CAMEO groups: 3: Acids, Carboxylic H-codes: H315, H318, H372 GHS pictograms: corrosive, health-hazard PubChem flash point: 121.1 °C PubChem vapour pressure: 0.1 kPa PubChem pH: 3.4 PubChem physical form: solid PubChem physical description: Benzoic acid appears as a white crystalline solid. Slightly soluble in water. The primary hazard is the potential for en	30 min weekly inhalation, skin	 

#	CHEMICAL	FORM / QTY	HAZARD STATEMENTS	WEL	STORAGE	EXPOSURE	GHS
3	Nitric acid CAS 7697-37-2	Forms: liquid Mass/volume range: 100 mL	H272 May intensify fire; H314 Causes severe skin burns and eye damage; H330 Fatal if inhaled	TWA: n/a STEL: 1 ppm (2.6 mg/m ³) Source: EH40	Oxidizing acids in double containment double containment; isolate from organics, flammables and bases; retain toxic-label controls and secondary containment; water-reactive - keep dry/review Source: GHS/PubChem + CAMEO Confidence: High CAMEO strong oxidizing acid group. CAMEO groups: 2: Acids, Strong Oxidizing H-codes: H272, H314, H330 GHS pictograms: oxidising, corrosive, toxic PubChem vapour pressure: 6.4 kPa PubChem physical form: liquid PubChem physical description: Nitric acid, other than red fuming appears as a pale yellow to reddish brown liquid with reddish brown vapors and a suff	15 min weekly inhalation, skin	

B1 APPENDIX B — COSHH ESSENTIALS EXPLANATION & SCREENING

COSHH Essentials is an HSE control-banding screen. LabCAT presents it as substance-level screening: it estimates a control approach for each substance from health hazard group, quantity scale, and volatility or dustiness. It is not a legal approval by itself: a competent assessor must confirm the output against SDS information, exposure route, quantity, duration, WELs and local conditions.

GUIDANCE: COSHH ESSENTIALS REFERENCE TABLES

APPROACH 1	General ventilation and good practice may be sufficient only when confirmed by a competent assessor for the actual task.
APPROACH 2	Engineering control is indicated, normally LEV or equivalent capture/control.
APPROACH 3	Containment or enclosure is indicated where small breaches may occur.
APPROACH 4	Specialist advice is required; do not rely on the banded screening alone.

HAZARD GROUP	MEANING
GROUP A	Lower health hazard band: H304, H315, H319, H336, EU66 and similar lower-toxicity effects.
GROUP B	Harmful / STOT category 2 band: H302, H312, H332, H371.
GROUP C	Toxic, corrosive, serious eye damage, sensitising skin, respiratory irritation or STOT category 1/2: H301, H311, H314, H317, H318, H331, H335, H370, H373, EU71.
GROUP D	Fatal acute toxicity, suspected carcinogen / reproductive toxicant, lactation hazard or repeated-exposure organ damage: H300, H310, H330, H351, H360, H361, H362, H372.
GROUP E	Respiratory sensitiser, mutagen, carcinogen category 1 / suspected mutagen or toxic by eye contact: H334, H340, H341, H350, EU70. Specialist advice is required (Approach 4).

EP BAND	MEANING
EP1	Lowest exposure potential: small amounts with low/medium dustiness, or low-volatility millilitre-scale liquids.
EP2	Low-to-moderate exposure potential: grams of high-dust solids, kg/tonne of low-dust solids, or millilitres of medium/high-volatility liquids.
EP3	Moderate-to-high exposure potential: litre-scale liquid work or kilogram-scale medium/high-dust solid work.
EP4	Highest exposure potential: cubic-metre-scale high-volatility liquids or tonne-scale medium/high-dust solids.

PREDICTED EXPOSURE RANGES BY EP BAND AND CONTROL APPROACH

This reference table shows the predicted airborne exposure ranges associated with each exposure predictor (EP) band and control approach. It helps explain why COSHH Essentials may move from general ventilation to engineering control or containment as the expected dust or vapour exposure increases. It is background guidance only: LabCAT still bases the recommendation for each substance on hazard group, quantity scale and volatility or dustiness, and the assessor must confirm the final controls against the task and SDS.

PREDICTED EXPOSURES FOR DUST IN AIR (MG/M3)			
EP BAND	CONTROL APPROACH 1	CONTROL APPROACH 2	CONTROL APPROACH 3
EP1 SOLID	0.01 to 0.1	0.001 to 0.01	<0.001
EP2 SOLID	0.1 to 1	0.01 to 0.1	0.001 to 0.01
EP3 SOLID	1 to 10	0.1 to 1	0.01 to 0.1
EP4 SOLID	>10	1 to 10	0.1 to 1

PREDICTED EXPOSURES FOR VAPOUR IN AIR (PPM)			
EP BAND	CONTROL APPROACH 1	CONTROL APPROACH 2	CONTROL APPROACH 3
EP1 LIQUID	<5	<0.5	<0.05
EP2 LIQUID	5 to 50	0.5 to 5	0.05 to 0.5
EP3 LIQUID	50 to 500	5 to 50	0.5 to 5
EP4 LIQUID	>500	5 to 500	0.5 to 5

RECOMMENDATIONS: SUBSTANCE-LEVEL SCREENING OUTPUT

The table below is LabCAT's substance-level COSHH Essentials screening result for this assessment. Each substance has its own approach; the highest approach present is highlighted for assessor review, but task-specific controls must still be confirmed.

SUBSTANCE	GROUP	H-CODES	SCALE	BAND	EP	APPROACH
Ethanol	A	—	small	medium (volatility)	EP2	Approach 1 — General ventilation
Benzoic acid	D	H372	small	unknown (dustiness)	EP1	Approach 2 — Engineering controls (LEV)
Nitric acid	D	H330	small	medium (volatility)	EP2	Approach 3 — Containment / enclosure